[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, VERTICAL, RAM AND TURRET, KNEE, MANUAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers manual, knee, ram and turret, vertical milling machines. The machines are intended for shaping workpieces by the material removal processes of drilling, reaming, tapping, milling, and boring.

2. CLASSIFICATION. The milling machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 7.2(b)). Milling machine sizes are shown in table 1.

Class 1 - 2-horsepower spindle motor rating
Class 2 - 3-horsepower spindle motor rating
Class 3 - 4-horsepower spindle motor rating
Class 4 - 5-horsepower spindle motor rating

Size - Sizes are shown in table I.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The milling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Identification of machine axes. The X-, Y-, and Z-axes of movement of the milling machine shall be identified in accordance with Electronic Industries Association (EIA) standard 267, "Axis and Motion Nomenclature for Numerically Controlled Machines".

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

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TABLE I. Sizes of milling machines.

Size	Class	Minimum worktable travel, inches		
		Longitudinal, X-axis	Transverse, Z-axis	Vertical, Y-axis
1	1	18	7	12
2		28	12	16
3		35	12	16
4		37	16	16
5	2	28	12	16
6		34	12	16
7		36	12	15
8		38	16	15
9		39	12	18
10	3	34	15	15
11		38	16	15
12	4	30	16	16
13		36	18	16
14		39	12	17

3.2 Components. The milling machine shall include, as a minimum, the following components:

3.2.1 Column and base. The column and base shall support the other elements of the milling machine and shall incorporate materials, shape, and mass to minimize adverse vibration. The base shall have provisions for leveling and anchoring the machine.

3.2.2 Knee. The knee shall be fitted to the column machineways and shall have machineways to support the saddle. The knee shall have vertical (Y-axis) traverse by means of an elevating screw operated by a crank or handwheel. If required, power feed to the knee shall be furnished as specified (see 7.2(c)).

3.2.3 Saddle. The saddle shall have machineways to support the worktable. The saddle shall have transverse (Z-axis) traverse by means of a screw operated by a crank or handwheel. If required, power feed to the saddle shall be furnished as specified (see 7.2(d)).

3.2.4 Machineways. The milling machine shall be equipped with wipers or other means for preventing dust and chips from entering the knee and saddle machineways.

3.2.5 Worktable. Unless otherwise specified (see 7.2(e)), the worktable length and width and weight-holding capacity shall be that considered by the manufacturer as standard. Unless otherwise specified (7.2(f)), the worktable shall have the number, size, and spacing of T-slots that are considered by the manufacturer as standard. The dimensions of the T-slots provided shall

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conform to the ANSI standard B5.1M, “Bolts, Nuts and Tongues, T-Slot, Their - 75”. The worktable shall have longitudinal (X-axis) traverse by means of a screw operated by a crank or handwheel. If required, power feed to the worktable shall be furnished as specified (see 7.2(g)).

3.2.6 Spindle head assembly. Unless otherwise specified (see 7.2(h)), the spindle quill shall have power feed at three feed rates, adjustable automatic feed disengagement in both directions, and a micrometer depth stop mechanism. Unless otherwise specified (see 7.2(i)), the spindle motor shall provide variable spindle speeds in accordance with the manufacturer’s standard machine design. Unless otherwise specified (see 7.2(j)), the spindle assembly shall swivel 90 degrees right and left from vertical and 45 degrees front and back from vertical. Unless otherwise specified (7.2(k)), the spindle taper shall be that considered by the manufacturer as standard. If required, a power drawbar shall be furnished as specified (see 7.2(l)).

3.2.7 Ram. Unless otherwise specified (see 7.2(m)), the milling machine shall have an overarm ram which moves the spindle assembly a minimum of 12 inches horizontally across the worktable. Ram positioning shall be by manual means.

3.2.8 Turret. The milling machine shall have a turret that allows the ram and spindle assembly to rotate 360 degrees around the vertical axis of the column by manual means.

3.2.9 Locking devices. The milling machine shall be equipped with locking or clamping devices to hold the ram, turret, spindle assembly, table, saddle, and knee at any position in their range of operation.

3.2.10 Dials and indicators. Unless otherwise specified (see 7.2(n)), angular scales shall be divided into 1-degree, or smaller, increments and linear scales into 0.001-inch or 0.02-millimeter increments. The zero-degree position of angular scales shall be prominently marked.

3.2.11 Controls. The milling machine shall have manually operated handwheels, switches, locks, and levers. If required, digital readout and control systems for worktable traverses shall be furnished as specified (7.2(o)).

3.2.12 Lubrication system. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible, and a manually operated “one shot” lubrication system shall be provided to deliver oil to the table, saddle, and knee machineways and leadscrews. A sight gauge, warning light, or similar device shall be provided for determining lubricant level in the reservoir(s). If required, an automatic lubrication system shall be provided as specified (see 7.2(p)).

3.2.13 Coolant system. Unless otherwise specified (see 7.2(q)), the milling machine shall have a mist coolant system. A means shall be provided to direct and control the amount of coolant to the cutting tool over the entire work area. A base- or floor-mounted splash tray shall be provided to catch coolant overflow.

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3.2.14 Electrical system. Unless otherwise specified (see 7.2(r)), the electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) standard 79, "Electrical Standard for Industrial Machinery". The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(s)). Unless otherwise specified (see 7.2(t)), the control system shall operate on a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.15 Electric motor. The milling machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers (NEMA) standard MG-1, "Motors and Generators". The drive motor horsepower is a continuous duty rating. Unless otherwise specified (see 7.2(u)), the motors shall have open dripproof enclosures.

3.2.16 Optional accessories. Optional accessories shall be provided as specified (see 7.2(v)).

3.3 Performance. Unless otherwise specified (see 7.2(w)), the milling machine shall meet the National Aerospace Standard (NAS) standard 985, "Uniform Alignment Tests", alignment requirements shown in table II. Unless otherwise specified (see 7.2(x)), the machine shall meet the performance requirements of sections 5.3.1 through 5.3.4.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(y)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI standard B11.8, "Safety Requirements for Construction, Care, and Use of Drilling, Milling, and Boring Machines and Machine Tools". The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(z)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(aa)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

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TABLE II. Alignment requirements.

NAS 985 Uniform Alignment Tests	
Paragraph	Alignment Parameter
4.3.2.1	Flatness - work mounting surface
4.3.3.1	Rise and fall of work mounting surface
4.3.4.1	T-slots parallel with longitudinal movement
4.3.5.1	Transverse movement square with center T-slot
4.3.6.1	Center T-slot square with spindle (vertical)
4.3.7.1	Head movement square with work mounting surface
4.3.8.1	Work mounting surface square with spindle
4.3.9.1	Transverse movement square with longitudinal movement
4.3.10.1	Spindle runout
4.3.13.1	Alignment of spindle with arbor support (if provided)
4.3.18.1	Work mounting surface square with vertical knee movement
4.3.21.1	Deflection of table under load
4.3.22.1	Slide displacement due to clamping

3.7 Nameplate. A nameplate shall be attached to the machine. Unless otherwise specified (see 7.2(bb)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(cc)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4.2 Environmental protection. The item shall meet applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description; shall conform to the producer's own drawings, specifications, standards, and quality assurance practices; and shall be the same product as that which the supplier offers for sale on the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(dd)), the machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(ee)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the government. The manufacturer shall provide test instruments and equipment with calibration traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(ff)). Unless otherwise specified (7.2(gg)), the tests shall be performed indoors with ambient conditions of 40 to 105 degrees Fahrenheit and 20- to 90-percent humidity.

5.3.2 Operational test. The machine shall be operated in accordance with the manufacturer's standard operating procedures for initial warm-up and run-off checks. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The following performance tests shall be conducted without making adjustments to or performing any maintenance on the machine.

5.3.3.1 Alignment tests. The alignment test results shall meet the requirements shown in table II.

5.3.3.2 Horsepower, milling accuracy, and tool chatter tests. The milling machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall not be less than 16 inches long, 6 inches wide, and 3 inches thick and may be drilled to facilitate clamping. One large surface shall be ground flat and mounted in contact with the table. Roughing cuts shall be made on the horizontal surface and on two adjacent vertical surfaces of the workpiece at sufficient depth and feed rate to load the spindle drive motor to at least the maximum duty rated electrical input. The type of cutting tool, depth of cut, spindle speed, and feed rate shall be determined by

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the manufacturer, provided the required loading is achieved. The resultant metal removal rate shall be at least 0.65 cubic inches per minute rated horsepower of the spindle drive motor. There shall be no evidence of tool chatter, motor stalling, or permanent distortion of the workpiece. Finishing cuts shall be made on the horizontal and vertical surfaces and shall meet the accuracy stated in 5.3.3.3 through 5.3.3.5.

5.3.3.3 Squareness. The two milled sides of the workpiece shall be measured for squareness. The sides shall be square with each other within 0.001 inch per foot.

5.3.3.4 Flatness. The two milled sides of the workpiece shall be measured for flatness. The sides shall be flat within 0.001 inch per foot.

5.3.3.5 Parallelism. The finished horizontal surface of the workpiece shall be measured for parallelism. The top surface shall be parallel to the bottom surface within 0.001 inch per foot.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(hh)).

5.4 Acceptance. Unless otherwise specified (see 7.2(ii)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment tests shall be compared with the alignment requirements shown in table II. Failure of the machine to meet the alignment requirements shown in table II shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified (see 7.2(jj)), the products shall be preserved, packed, and marked as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 EIA standards. Copies of EIA standards may be obtained from the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3824.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 NAS standards. Copies of NAS standards may be obtained from the National Standards Association, Incorporated, 1321 Fourteenth Street NW, Washington, DC 20005.

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7.1.6 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.1.7 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Class and size of machine required (see 2)
c. Knee power feed, as specified (see 3.2.2)
d. Saddle power feed, as specified (see 3.2.3)
e. Worktable length, width, and capacity, as specified (see 3.2.5)
f. Number, size, and spacing of worktable T-slots, if different (see 3.2.5)
g. Worktable power feed, as specified (see 3.2.5)
h. Spindle quill power feed, if different (see 3.2.6)
i. Spindle variable speeds, if different (see 3.2.6)
j. Spindle head swivel, if different (see 3.2.6)
k. Spindle taper size, if different (see 3.2.6)
l. Power drawbar, as specified (see 3.2.6)
m. Overarm ram, if different (see 3.2.7)
n. Angular and linear scale incremental divisions, if different (see 3.2.10)
o. Digital readout and control system, as specified (see 3.2.11)
p. Automatic lubrication system, as specified (see 3.2.12)
q. Mist coolant system, if different (see 3.2.13)
r. Electrical system industrial standard, if different (see 3.2.14)
s. Electrical system voltage, as specified (see 3.2.14)
t. Control system voltage, if different (see 3.2.14)
u. Motor enclosures, if different (see 3.2.15)
v. Optional accessories, as specified (see 3.2.16)
w. Alignment requirements, if different (see 3.3)
x. Performance requirements, if different (see 3.3)
y. Machine dimensions and weight, as specified (see 3.4)
z. System of units required (see 3.6)
aa. Dual system of units, if required (see 3.6)
bb. Nameplate, if different (see 3.7)
cc. Lubrication plate or chart, if required (see 3.8)
dd. Testing, if different (see 5.3)
ee. Test responsibility, if different (see 5.3.1)
ff. Certification of calibration, as specified (see 5.3.1)
gg. Environmental conditions, if different (see 5.3.1)
hh. Optional tests, as specified (see 5.3.4)
ii. Acceptance, if different (see 5.4)
jj. Packaging requirements, if different (see 6.1)

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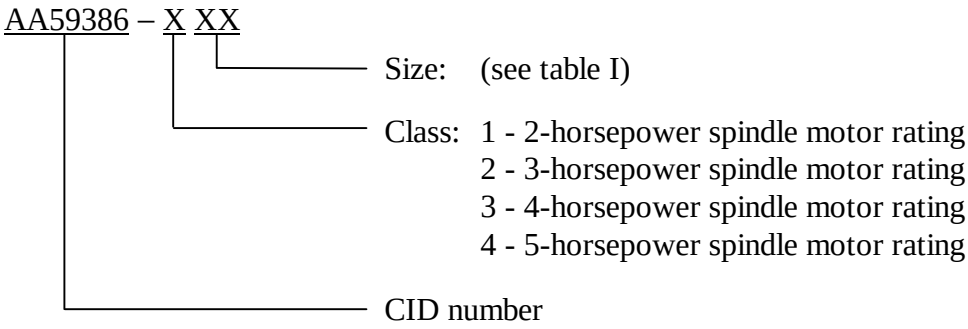
7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Navy - MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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